WEEK – 1

Minor Project (BTCS607N)-Weekly Progress Report

Project Title: Drone-Based Intelligent System for Apple Orchard Management in Himachal Pradesh.

Group ID: G9

Branch & Section: BTech CSE, AIDS -A (AIML with Microsoft)

Date: 22-01-2026

1. Executive Summary

- **Current Sprint:** Sprint 1(Issue 1 project understanding and planning)
- **Sprint Goal:** To define the problem scope, perform requirement analysis, and set up project management using Jira.
- **Status:** On Track

2. UML & OOAD Progress

- **Diagrams Completed this week:** High-level System Architecture Diagram (conceptual),

  Initial Use Case identification (for orchard monitoring, yield estimation, and pest management).
- **Diagrams Revised:** None (initial planning phase)
- **OO Concept Applied:** Identification of core system objects such as *Drone*, *Image Processor*, *AI Model*, *Farmer Dashboard*

  Applied **abstraction** to separate data collection, processing, and decision-making modules.

3. Technical Execution (DevOps & Git)

- **Jira Tasks Completed:** Project Research & Literature Review, Requirement Analysis, System Architecture Design
  (Tracked using Jira backlog and Sprint 1 tasks)

Link: https://ayushshrivastava.atlassian.net/jira/software/projects/DBISFAOM/summary

- **Git Activity:**
  - Active Branches: main

- Merged Pull Requests: Not applicable (initial planning phase)

Link : https://github.com/yp1818/Drone-Based-Apple-Orchard-Management

- **CI/CD Status:** Not configured yet (planned for upcoming development sprint)

4. Testing Summary

- **Total Unit Tests:** 0 (Planning phase)
- **Pass Rate:** Not applicable
- **Key Bug Identified:** Understanding and organizing Jira Scrum workflow for academic project tracking was initially challenging and was resolved through proper backlog and sprint setup.

5. Goals for Next Week

- Design and finalize **UML diagrams** (Use Case Diagram, Class Diagram, and Sequence Diagram) for the Drone-Based Orchard Management System.
- Plan and document the **data flow and AI modules**, including image processing workflow and tree health & yield estimation logic.

Mentor Signature **Project Guide Signature**